

SIMULTANEOUS GO VIA QUANTUM COLLAPSE

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Abstract. We give one definition of simultaneous games and simultaneization of a deterministic game with two players. As the main motivation and example, we construct a simultaneous game SGo, which is a simultaneization of Go. SGo is many ways a simpler game than Go, as Komi and Ko rules are removed. On the other hand the states are more complex, because there are “entanglement phenomena”, an idea inspired by quantum mechanics. But instead of a dice roll there is a certain deterministic “quantum state” reduction, so that state evolution is deterministic. It is an apparently open question whether SGo has a draw strategy.

1. Introduction

The game Go is an ancient and interesting game of local to global geometric principles. Like all turn based games it suffers from time asymmetry between the players Black and White. In Go this is addressed by a somewhat ad-hock rule called komi: whereby the player White which moves second is given a specific point advantage. The trouble of time asymmetry goes deeper. For Go, it results in counter-intuitive local pathologies in game states, for example Ko situation, where after a capture of a stone the opposing player may immediately recapture, but is forbidden by an extra ad-hock rule, to avoid an infinite loop situation.

At the same time, precisely because of its local to global nature Go is “almost simultaneous”. We make this mathematically precise by introducing one notion of simultaneous games and show that Go has a simultaneization that we call SGo. The latter is a deterministic game, which is totally playable on a standard Go board, although we need a third type of stones that we call red. SGo loses the Komi and Ko rules, but has a somewhat more complicated capture rule because the game states are more complex. The idea for SGo is inspired by quantum mechanics. Probably the most interesting question at the moment is whether SGo has a strategy to draw. We partly discuss this at the end of the paper.

Some fundamental advances in machine learning recently lead to computer Go agents winning against stellar human opponents, particular alphaGo winning against Lee Sedol. At the same time computer agents do much worse in games with imperfect information, like Poker.¹ It is interesting to speculate how alphaGo and its cousins would do against a human opponent in SGo, which is both more complex in its state space and is an imperfect information game.

2. Game rules

The game is initialized with an empty standard Go board, with virtual horizontal/vertical lines whose intersections we just call intersections. There are two

¹At least in Poker it seems that computer agents have caught up as of 2025.

players that we call Black and White. As in Go, players Black and White move on a given turn by placing a black respectively white stone on an empty intersection on the board, but in SGo they do this on the same turn without knowledge of the others move on this turn. This requires modifying the placement and capture rules of Go, while removing the Komi rule, and Ko rules. We also introduce a third type of stones called red.

2.1. Some definitions. Given a board game state S , that is any collection of white, black, red stones on the board let $G(S)$ denote the graph with vertices stones, and edges between vertices whenever the corresponding stones are horizontally or vertically adjacent. (Meaning they are adjacent and on the same horizontal/vertical line.) From now on adjacency, always means this type of adjacency. In what follows the terms stones and vertices may be used synonymously.

A colored component is a connected component of the full sub-graph of $G(S)$ whose vertices are either all the black, or all the white stones. We call this a white, respectively black component. (Full sub-graph means that the inclusion map of the sub-graph induces a bijection on sets of edges.)

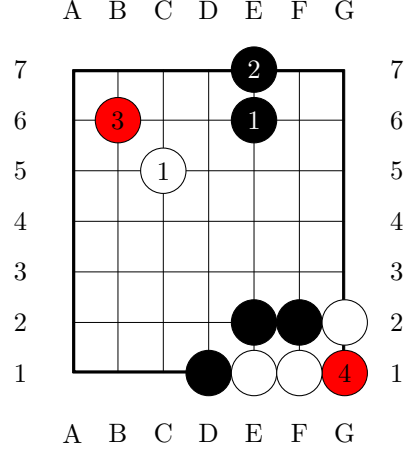
As in classical Go, a connected component C has no liberties if it cannot be extended (as a connected component) by placing a stone on the board. We will also call C a trapped component. We define stones adjacent to C , as those stones whose corresponding vertices are connected by an edge to C in $G(S)$.

Given a colored component C , A white/black stone placed at an intersection is called a trapping if it is adjacent to a trapped colored component.

2.2. Stone placement resolution rule of SGo. If the players move to the same intersection on the board we place a stone there whose color resolves according to the following. If either a white or black stone placed at the intersection is trapping, then a red stone is placed at the intersection. We call this a trapping move. Otherwise, if the move is not a trapping move we have the following:

- (1) If the intersection is adjacent to a black stone, and neither a white stone nor red stone, a black stone is placed at the intersection.
- (2) If the intersection is adjacent to a white stone, and neither a white stone nor red stone a white stone is placed at the intersection.
- (3) Otherwise, a red stone is placed.

To the right is an example of placement resolution. Note that the stone placed on move 4 does not resolve to white, as it is a trapping move.



2.3. Capture rule of SGo. After both White and Black move on current turn numbered n , using the stone placement rule above, we get some board state S . The stage 1 capture rule is then invoked on this turn as follows. Let C be a colored component with no liberties of the board state S , (the stones of C may have been placed on this turn.) Suppose that:

- (1) There is no red stone adjacent to C placed on turn n .
- (2) Either some stone adjacent to C was placed on turn n , or some stone in C was placed on turn n .

Then we call C trapped on turn n , the stones of C are also called trapped on turn n turn. Suppose that no stone adjacent to C was trapped on this turn. Then C is resolved as captured. To paraphrase, if C has no liberties, and any trapped respectively red stones adjacent to C , were trapped respectively created on some previous move, then C is resolved as captured.

In the case C is captured, supposing C is white:

- The stones of C are removed as prisoners of Black as in Go.
- Any red stones adjacent to C resolve as black. (Concretely are replaced by black stones.)

If C is black the process is reflected. The above stage 1 capture rule is invoked recursively until no captured colored components remain. We call this stage one reduction and the resulting board state will be denoted by $R_n(S)$.

Lemma 1. The stage one reduction is well defined, that is independent of the order in which killed components are captured.

Proof. Suppose we have two colored components C_0, C_1 captured on the given turn. By assumptions no stones in C_1 can be adjacent to stones of C_0 , and vice versa. Hence applying capture rule to the group C_0 and then to group C_1 , results in the same board state as first applying the capture rule to the group C_1 and then the group C_0 . The result clearly follows. $\square$

If there are no colored components trapped on the turn $n - 1$ or lower, the turn is resolved. Otherwise, we recursively define higher stage reduction as follows.

Define stage k reduction by the condition that the board state after stage k reduction denoted by $R^k(S)$ satisfies:

$$\begin{aligned} R^1(S) &= R_n(S). \\ R^k(S) &= R_{n-k+1}(R^{k-1}(S)). \end{aligned}$$

If there are no colored components trapped on the turn $n - k$ or lower, the turn is resolved after stage k reduction.

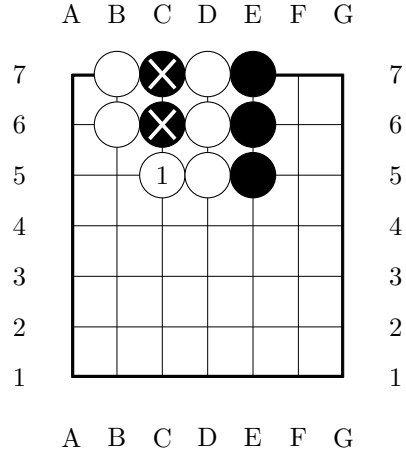
Remark 2. This may appear to be fairly complicated, but in practice the resolution of most turns is easy to work out mentally, we will give some examples of such “nested entanglement states” further ahead. Of course having a computer work out turn resolution automatically is very helpful.

2.4. End of the game. As in Go, SGo game ends after both players pass. At which point the winner is decided as in Go by territory and prisoner count. White’s territory boundary consists of connected components white/red stones. Black’s territory boundary consists of connected components of black/red stones. Red stones are never counted as prisoners. A draw is now possible although compared to chess it might be much less likely, (on a typical 13×13 or 19×19 go board). See Section 3.2 for an example.

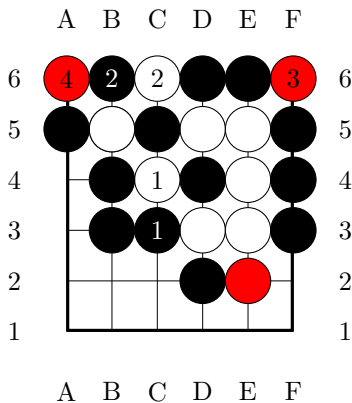
3. Examples

We give here some examples. In these example, the players need only play the position and not attempt to predict the moves of the other player, at higher level of play and more marginal, complex positions there may be such mind games.

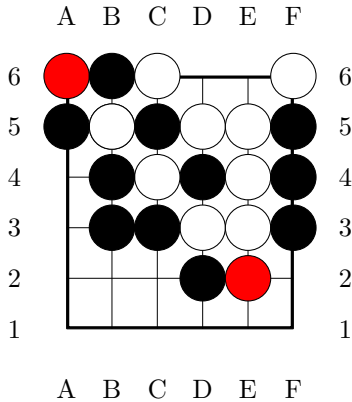
Black’s left group cannot be saved unless White allows it. On move 1 both Black and White move to C5, the stone on C5 resolves to white.



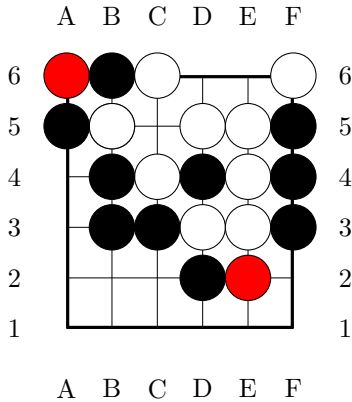
In the position on the right, on turn 1, White attempts capture at C4, while Black attempts capture at C3. This gives an example of a “nested entangled state”. Here is a possible continuation.



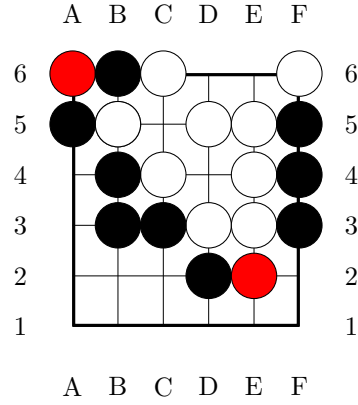
At the end of turn 4, stage one resolution is depicted on the right.



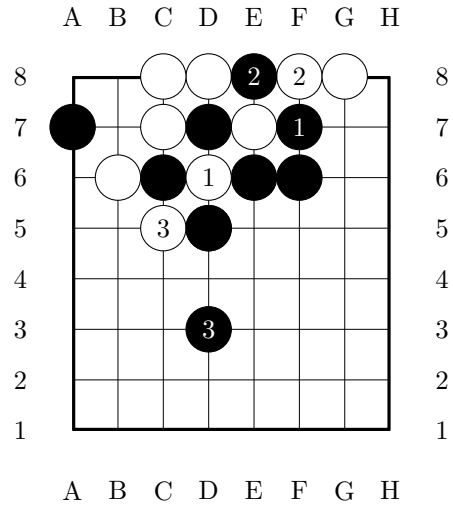
Stage two resolution is depicted on the right.



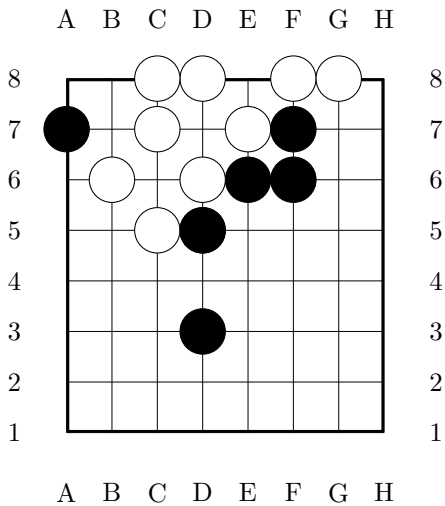
Stage three and final state resolution is depicted on the right.



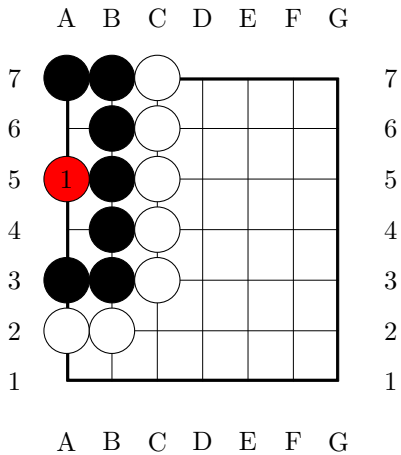
Ko situation. Recall, that we don't have a Ko rule, but it is now clearly unnecessary as we can't have any infinite loops. See example on the right. By the capture rule White's d5 is not a capture, instead it creates an "entangled state". This position is not great for White, but there is a "mind game" here. On turn 2 White predicts Black's move, and further entangles the position. White now has chances, a possible continuation is shown.



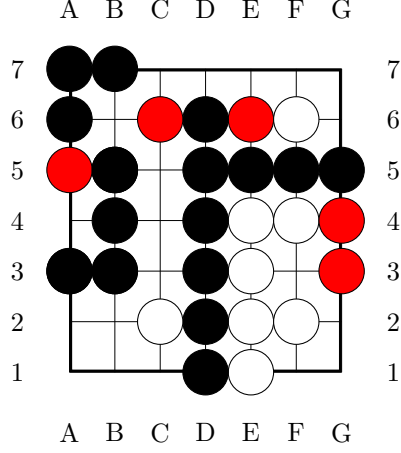
Resolution after turn 3.



3.1. Life and death. Life and death is positionally very similar to Go, here is an example.



3.2. End game position. Both players pass and the game ends. Assuming there are no prisoners. White has 4 points, Black has 19 points.



4. SGo as a simultaneazation of Go

We give here a general definition of simultaneazation of a deterministic game, and show that *SGo* is a simultaneazation of *Go*. The result is not unsurprising, for example chess likely does not have a simultaneazation in our sense, although formal verification of this is not obvious. First we give some basic definitions. We will treat (finite) deterministic games as certain types of (finite) deterministic automata.

Definition 1. A (finite) deterministic game G with 2 players P_0, P_1 or Black, White, consists of:

- A (finite) deterministic automaton G , whose set of states is a set $S(G)$ called the set of game states of G .
- Nonempty subsets $W_0(G), W_1(G), D(G) \subset S(G)$ of final (accepting) states with empty pairwise intersection, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in S(G)$, understood as the state in which the game is initialized.
- A $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \sqcup M_1(G)$: interpreted as possible moves of Black and White.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \rightarrow S(G).$$

We recursively define:

$$E(s, m_1, \dots, m_k) = E(E(s, m_1, \dots, m_{k-1}), m_k).$$

Definition 2. A deterministic game G is called simultaneous if the following conditions are satisfied:

- (1) The set $S(G)$ has a distinguished representation as

$$S(G) = B(G) \sqcup B(G) \times M_0(G) \sqcup B(G) \times M_1(G),$$

where $B(G)$ is some finite set. ($B(G)$ may be interpreted as the set of “board states”).

- (2) The function E has the properties: if $s \in B(G) \subset S(G)$, then $E(s, m) = (s, m)$ for all $m \in M(G)$. If $s \in B(G) \times M_0(G)$, and $m \in M_1(G)$ then $E(s, m) \in B(G)$ (or undefined). If $s \in B(G) \times M_1(G)$, and $m \in M_0(G)$ then $E(s, m) \in B(G)$ (or undefined). E is undefined in all other cases.
- (3) Suppose that $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, $s \in B(G)$, $E(s, m_0)$ is defined and $E(s, m_1)$ is defined then $E(s, m_0, m_1)$ is defined.
- (4) For s, m_0, m_1 as in the previous property, if $E(s, m_0, m_1)$ is defined then so is $E(s, m_1, m_0)$ and

$$E(s, m_0, m_1) = E(s, m_1, m_0).$$

Note that if G is simultaneous by definition above it merely means that its state evolution rules allow it to be played with simultaneously played moves. But it may also be played with White and Black taking turns, and also knowing each other's move.

Definition 3. Let G_0, G_1 be deterministic games, with G_1 simultaneous. We say that G_1 is a simultaneization of G_0 if the following is satisfied.

- (1) $M(G_1) = M(G_0)$ as $\mathbb{Z}_2$ graded sets.
- (2) There is a map $\phi : S(G_1) \rightarrow S(G_0)$ s.t. $E_{G_1}(s, m)$ is defined if and only if $E_{G_0}(\phi(s), m)$ is defined.

Theorem 1. SGo is a simultaneization of Go.

Proof. We first outline the formal description $G = SGo$ as a simultaneous game, based on the description in Section 2. The space $M(G)$ is the same as Go, we suppose that $M_0(G)$ corresponds to moves of Black. The state space is:

$$S(G) = B(G) \sqcup B(G) \times M_0(G) \sqcup B(G) \times M_1(G),$$

where $B(G)$ is the set of all legal board positions as in the description in Section 2, meaning that the position is already resolved with respect to the movement resolution, and capture rule of SGo.

The evolution map E is then obviously defined. For example if $b \in B(G)$ and $m_0 \in M_0(G)$, $m_1 \in M_1(G)$ are legal moves then $E((b, m_0), m_1) \in B(G)$ is defined to be the resolution of the board position, using the capture rule of Section 2.3, after Black's move m_0 , and White's move m_1 are reflected on the board.

Define $\phi : S(G) \rightarrow S(G_0)$ by the following conditions. If $s \in B(G)$ then $\phi(s)$ is the Go board state obtained by replacing every stone on the board by a black stone. If $s \in B(G) \times M_i(G)$ then $\phi(s) = \phi(b)$, where $b = pr(s)$, for $pr : B(G) \times M_i(G) \rightarrow B(G)$ the projection. It is immediate from construction of SGo that Property 2 is satisfied. $\square$

4.1. Solvability. A theorem of Zermelo [1] says roughly that in every deterministic game which is strictly turn based (we haven't really defined such a notion, but we could do so also in the language automata as above), either Black or White has a winning strategy or both have a draw strategy. Obviously in a simultaneous deterministic game as we defined, Black and White cannot have a winning strategy, as that would be absurd.

Question 1. When does a simultaneous game has a strategy to draw?

Some conditions must be necessary as rock-paper-scissors is a simultaneous game, and has no drawing strategy.

Question 2. Does SGo have a strategy to draw?

References

- [1] U. Schwalbe and P. Walker, Zermelo and the early history of game theory, <http://www.math.harvard.edu/~elkies/FS23j.03/zermelo.pdf>.